Analysis of Covid-19 chest x-rays

Data Science Project Report

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LIORA - Data Science

September 9, 2026

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Chapter 1

Introduction

1.1 Background and Motivation

COVID-19 is a respiratory disease caused by the SARS-CoV-2 virus. During the pandemic, rapid and reliable diagnosis was an important challenge. Chest X-ray analysis has been explored as a possible tool to support diagnosis because abnormalities in lung structures can appear in infected patients.

1.2 Problem Statement

This project investigates the automated classification of chest X-ray images into four diagnostic categories: COVID-19, Normal, Lung Opacity, and Viral Pneumonia. Chest X-ray imaging is widely used as a fast and accessible diagnostic tool for respiratory diseases, making automated image classification a valuable application of deep learning. The objective of this project is to develop and evaluate convolutional neural network models for multi-class classification while assessing their robustness and interpretability. The dataset consists of publicly available chest X-ray images collected from several repositories and includes corresponding lung segmentation masks. Details of the dataset and its characteristics are presented in Chapter 2.

The project focuses on image classification and does not aim to provide a medical diagnosis or replace healthcare professionals. Furthermore, the objective is not to localize abnormalities within the images.

Chapter 2

Dataset

2.1 Overview

The dataset consists of chest X-ray (CXR) images belonging to four diagnostic categories: *COVID-19*, *Normal*, *Lung Opacity*, and *Viral Pneumonia*. All images are provided in PNG format with a resolution of 299×299 pixels. In addition, lung segmentation masks are available for every image. The dataset was compiled from several publicly available repositories and published studies, with detailed source information provided in the accompanying metadata.

According to the dataset documentation, the four classes were collected from different source repositories rather than a single unified acquisition process. Table 2.1 summarizes the origin of each class.

Table 2.1: Sources of the chest X-ray images for each class.

Class	**Source(s)**
COVID-19	PadChest; German Medical School; SIRM/GitHub/Kaggle/Twitter; another GitHub source
Normal	RSNA Pneumonia Detection Challenge (8,851) + Kaggle Chest X-ray Pneumonia Dataset (1,341)
Lung Opacity	RSNA Pneumonia Detection Challenge only
Viral Pneumonia	Kaggle Chest X-ray Pneumonia Dataset only

The COVID-19 class was created by aggregating confirmed COVID-19 chest X-ray images from multiple public repositories to increase the number of available samples. Normal images were obtained primarily from the RSNA Pneumonia Detection Challenge dataset, with additional normal cases collected from the Kaggle Chest X-ray Pneumonia dataset. Lung Opacity images originate exclusively from the RSNA dataset and represent non-COVID pulmonary opacities. Viral Pneumonia images were obtained exclusively from the Kaggle Chest X-ray Pneumonia dataset.

Although combining multiple repositories increases the diversity of the dataset, it also introduces potential biases resulting from differences in image acquisition protocols, imaging equipment, preprocessing pipelines, compression methods, image quality, and patient populations. In particular, all Lung Opacity images originate from the RSNA dataset, whereas all Viral Pneumonia images are drawn exclusively from the Kaggle dataset. Consequently, a classifier may learn source-specific characteristics instead of disease-related radiographic patterns.

This issue is commonly referred to as *shortcut learning*, where a model exploits acquisition- or dataset-specific artifacts rather than clinically relevant pathological features. To ensure that the trained model focuses on meaningful anatomical regions, explainable artificial intelligence (XAI) techniques such as Grad-CAM should be employed to verify that predictions are based on disease-related radiographic features instead of scanner- or dataset-specific artifacts.

2.2 Dataset Composition and Class Balance

The dataset consists of 21,165 chest X-ray (CXR) images distributed across four diagnostic categories: *COVID-19*, *Lung Opacity*, *Normal*, and *Viral Pneumonia*. Each chest X-ray image has a corresponding lung segmentation mask, resulting in a one-to-one mapping between images and masks.

Table 2.2 summarizes the distribution of images across the four diagnostic classes. The dataset exhibits a noticeable class imbalance. The *Normal* class is the largest, accounting for 48.15% of all images, followed by *Lung Opacity* (28.41%), *COVID-19* (17.08%), and *Viral Pneumonia* (6.35%).

Table 2.2: Distribution of chest X-ray images across diagnostic classes.

Class	**Images**	**Percentage**
COVID-19	3,616	17.08%
Lung Opacity	6,012	28.41%
Normal	10,192	48.15%
Viral Pneumonia	1,345	6.35%
Total	**21,165**	**100.00%**

Figure 2.1 illustrates the class distribution within the dataset.

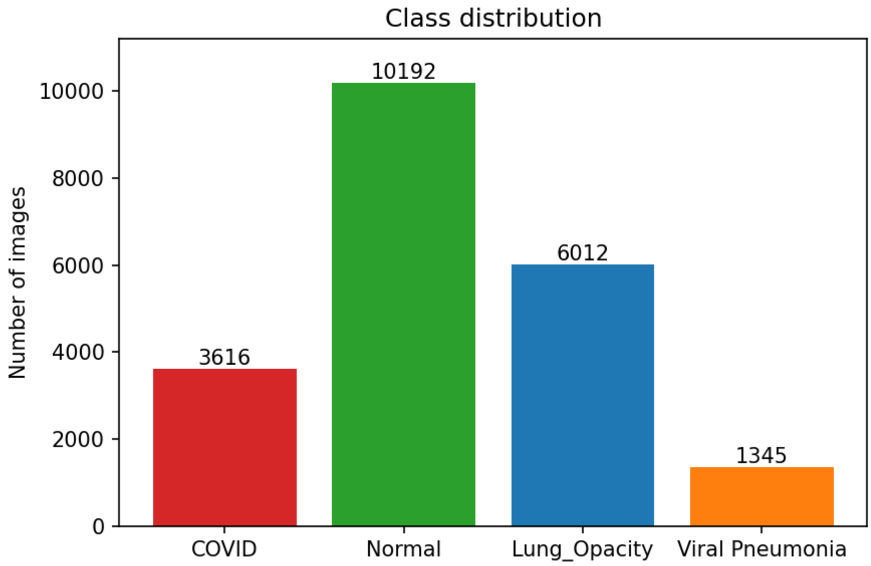


Figure 2.1: Distribution of chest X-ray images across the four diagnostic classes.

To determine whether the observed class frequencies differed significantly from an equal class distribution, a chi-square goodness-of-fit test was performed. The test indicated a highly significant deviation from uniformity ($\chi^2 = 8110.78$, $p < 0.001$). Furthermore, the effect size was large (Cohen's $w = 0.619$), confirming the presence of substantial class imbalance.

The observed imbalance has important implications for model development and evaluation. During dataset partitioning, stratified sampling should be employed to preserve the original class proportions across the training, validation, and test sets. Furthermore, overall classification accuracy alone may provide a misleading assessment of model performance. Instead, evaluation should include class-sensitive metrics such as per-class precision, recall, macro-averaged F1-score, balanced accuracy, and the confusion matrix to ensure that performance is assessed fairly across all diagnostic categories.

2.3 Structural Integrity, Dimensions, and Image Encoding

To verify the integrity of the dataset, all chest X-ray images and lung segmentation masks were examined prior to model development. Every one of the 21,165 chest X-ray images had a corresponding segmentation mask with an identical filename, indicating a complete one-to-one correspondence between images and masks. Furthermore, no unmatched image–mask pairs or unreadable PNG files were identified.

The average X-ray file size varied slightly across the four diagnostic classes. The mean file sizes were 35.00 KB for COVID-19, 34.40 KB for Lung Opacity, 36.39 KB for Normal, and 39.36 KB for Viral Pneumonia. The relatively larger mean file size observed for the Viral Pneumonia class is explained by the presence of RGB images within this class, which are substantially larger than their grayscale counterparts.

Table 2.3 summarizes the structural properties of the dataset.

Table 2.3: Structural properties of the chest X-ray images and segmentation masks.

Property	**X-rays**	**Masks**
Number of files	21,165	21,165
Dimensions	299×299 px	256×256 px
L mode	21,025	0
RGB mode	140	21,165

Although every image has a corresponding segmentation mask, the image and mask dimensions differ. All chest X-ray images have a resolution of 299×299 pixels, whereas all segmentation masks have a resolution of 256×256 pixels. Consequently, any pixel-wise operation involving both the X-ray image and its corresponding mask requires the mask to be resized or otherwise aligned with the image dimensions.

2.3.1 RGB Images in the Viral Pneumonia Class

A detailed inspection of image encoding revealed that all 140 RGB images are contained exclusively within the Viral Pneumonia class. These RGB images represent 10.41% of the Viral Pneumonia samples (140 of 1,345 images), while every image in the remaining three diagnostic classes is stored as a single-channel grayscale (L-mode) image. Consequently, image encoding is associated with the diagnostic class in the raw dataset.

Table 2.4 summarizes the characteristics of the Viral Pneumonia images according to image encoding.

Table 2.4: Comparison of grayscale (L) and RGB images within the Viral Pneumonia class.

Mode	**n**	**Mean (KB)**	**SD**	**Median**	**Range (KB)**
L	1,205	37.45	2.54	37.47	26.37–47.27
RGB	140	55.82	5.86	56.28	39.83–71.42

Figure 2.2 compares the mean PNG file sizes of grayscale and RGB images within the Viral Pneumonia class.

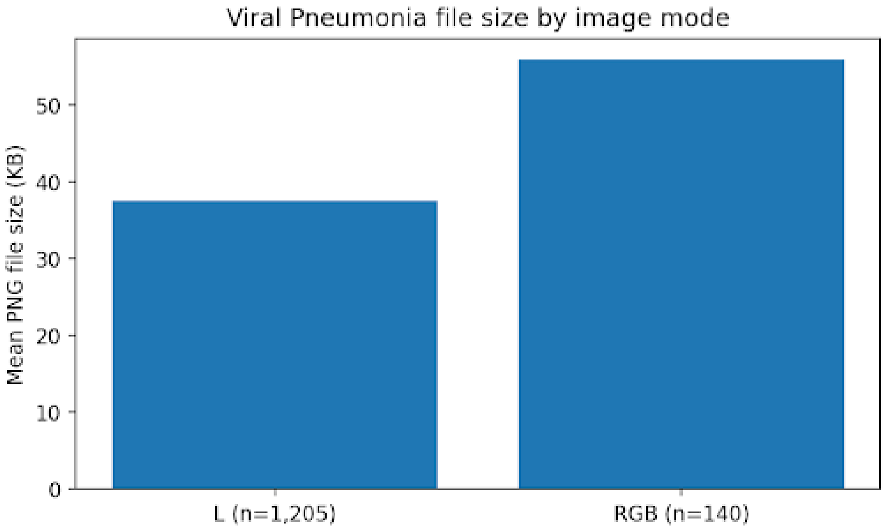


Figure 2.2: Mean PNG file size of Viral Pneumonia images stratified by image encoding.

As shown in Figure 2.2, RGB images are substantially larger than grayscale (L-mode) images within the same diagnostic category. This finding indicates that the elevated average file size observed for the Viral Pneumonia class is primarily attributable to differences in image encoding rather than disease-related image characteristics.

Since image encoding represents a technical characteristic rather than a clinically meaningful feature, all chest X-ray images should be converted to a common grayscale representation during preprocessing. Standardizing the image format eliminates a potential source of dataset bias and prevents the model from exploiting encoding-related differences as shortcut features.

2.4 Exact Duplicate Analysis

To identify redundant images, perceptual hashing (pHash) was applied using a Hamming-distance threshold of zero. This initial screening identified 287 candidate duplicate image pairs. A subsequent pixel-wise comparison confirmed that 91 of these pairs were exact pixel duplicates, whereas the remaining 196 pairs shared an identical perceptual hash but differed at the pixel level. All confirmed exact duplicates were found within the same diagnostic class, and no duplicate images were shared across different class labels.

Duplicate images were subsequently grouped into connected duplicate sets, and a single representative image was retained from each group. This process identified a total of 59 redundant files, corresponding to only 0.28% of the entire dataset. The distribution of redundant files across diagnostic classes is presented in Table 2.5.

Table 2.5: Redundant exact duplicate files identified after grouping connected duplicate sets.

Class	**Redundant Files**	**Percentage of Class**
COVID-19	51	1.41%
Lung Opacity	0	0.00%
Normal	1	0.01%
Viral Pneumonia	7	0.52%
Total	**59**	**0.28%**

Figure 2.3 illustrates the number of confirmed redundant images identified within each diagnostic class.

Figure 2.3: Distribution of confirmed redundant exact duplicate files by diagnostic class after grouping duplicate sets.

The identified duplicate images represent unnecessary redundancy within the dataset and, if retained, could introduce data leakage by allowing identical images to appear in both the training and evaluation sets. To prevent this issue, the 59 redundant files were removed prior to performing the training/validation/test split, while retaining one representative image from each exact-duplicate group. This preprocessing step helps ensure an unbiased evaluation of model performance and prevents overly optimistic performance estimates resulting from duplicate observations.

2.5 Whole-Image Brightness and Contrast

To characterize global image intensity, two descriptive features were calculated for each chest X-ray. *Brightness* was defined as the mean grayscale pixel intensity of the image, while *contrast* was represented by the standard deviation of grayscale pixel intensities within the image. To ensure consistency, all chest X-ray images were converted to grayscale prior to these calculations.

Table 2.6 summarizes the descriptive statistics for brightness and contrast across the four diagnostic classes. COVID-19 images exhibited the highest mean brightness (139.52), whereas Normal images showed the highest average global contrast (61.56). Lung Opacity and Viral Pneumonia displayed similar central brightness values, although their contrast differed slightly.

Table 2.6: Descriptive statistics of whole-image brightness and contrast by diagnostic class.

	Brightness			Contrast		
Class	**Mean**	**SD**	**Median**	**Mean**	**SD**	**Median**
COVID-19	139.52	25.03	142.59	54.65	14.83	54.05
Lung Opacity	126.02	23.57	120.82	57.40	9.81	56.97
Normal	129.39	22.45	125.51	61.56	9.55	63.53
Viral Pneumonia	125.37	18.98	126.86	58.75	10.78	59.71

Figure 2.4 presents the distributions of brightness and contrast for each diagnostic class. COVID-19 images tend to exhibit higher overall brightness, whereas the Normal class has the highest global contrast. The boxplots also reveal considerable within-class variability and numerous statistical outliers, particularly for COVID-19. Since these measurements were computed over the entire image, they may also be influenced by background regions outside the lungs. Consequently, evaluating brightness and contrast within the segmented lung regions may provide a more clinically meaningful assessment.

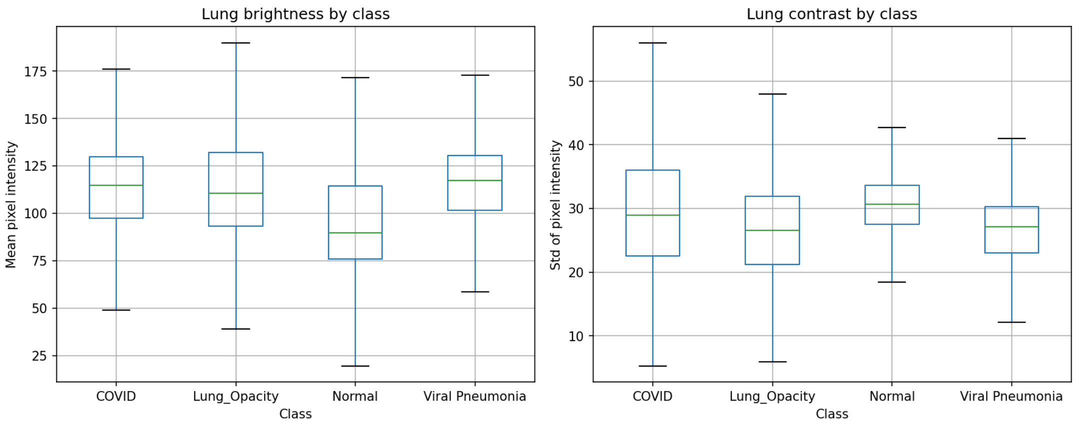


Figure 2.4: Whole-image brightness and contrast distributions by diagnostic class.

Figure 2.5 illustrates the relationship between whole-image brightness and contrast. Although class-specific tendencies are visible, substantial overlap exists among all four diagnostic classes. COVID-19 images exhibit a broader spread in both brightness and contrast, whereas Viral Pneumonia images are more tightly clustered. Normal images tend toward higher contrast, consistent with the descriptive statistics shown in Table 2.6. The considerable overlap indicates that global intensity characteristics alone are unlikely to provide sufficient discriminative information for reliable disease classification.

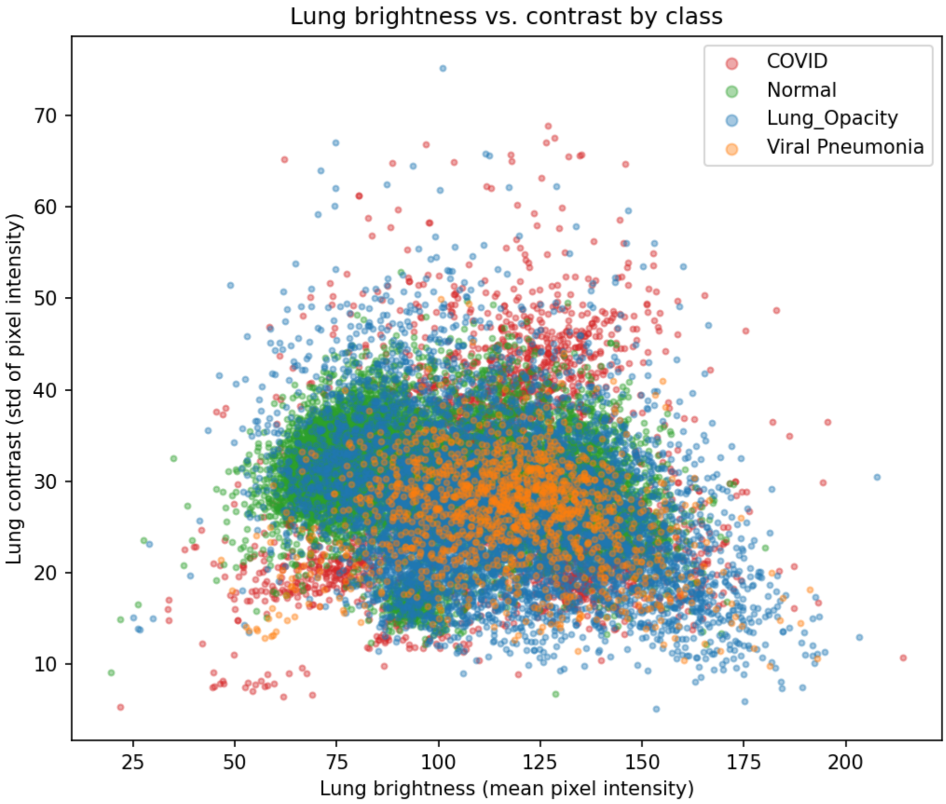


Figure 2.5: Relationship between whole-image brightness and contrast by diagnostic class.

The observed differences in global brightness and contrast are more likely to reflect variations in image acquisition protocols, scanner characteristics, and preprocessing procedures than disease-specific radiographic features. Consequently, these global intensity characteristics may represent technical confounding factors that a machine learning model could exploit as *shortcut features*. Applying intensity normalization during preprocessing and evaluating its impact on model performance can help determine whether the classifier relies on clinically meaningful radiographic patterns rather than global intensity differences.

2.5.1 Statistical Testing

Non-parametric statistical analyses were performed to compare brightness and contrast across the four diagnostic classes. A Kruskal–Wallis test indicated significant differences in whole-image brightness ($H(3) = 987.39$, $p < 0.001$), with a small overall effect size ($\varepsilon^2 = 0.0465$). Significant differences were also observed for whole-image contrast ($H(3) = 1393.78$, $p < 0.001$), with a modest effect size ($\varepsilon^2 = 0.0657$).

To quantify the magnitude of pairwise differences, Cohen's d was computed for each pair of diagnostic classes. The resulting effect sizes are summarized in Table 2.7.

Table 2.7: Pairwise Cohen's d effect sizes for whole-image brightness and contrast.

Comparison	**Brightness**	**Contrast**
COVID-19 vs Lung Opacity	0.559	-0.230
COVID-19 vs Normal	0.438	-0.618
COVID-19 vs Viral Pneumonia	0.601	-0.296
Lung Opacity vs Normal	-0.147	-0.431
Lung Opacity vs Viral Pneumonia	0.029	-0.135
Normal vs Viral Pneumonia	0.182	0.290

Post hoc Dunn tests with Bonferroni correction revealed significant brightness differences for every class pair except Lung Opacity versus Viral Pneumonia. Although the comparison between Normal and Viral Pneumonia remained statistically significant (adjusted $p = 0.0012$), the corresponding effect size was small ($d = 0.182$). For whole-image contrast, all six pairwise comparisons remained statistically significant following Bonferroni adjustment. The largest standardized difference was observed between COVID-19 and Normal ($d = -0.618$), indicating that COVID-19 images generally exhibit lower global contrast than Normal images.

Overall, these findings demonstrate that brightness and contrast differ systematically across diagnostic classes. However, the relatively small-to- moderate effect sizes and substantial overlap among class distributions suggest that global intensity characteristics alone are insufficient for robust disease classification. Instead, they should primarily be regarded as potential technical confounding variables that require careful preprocessing and interpretation during model development.

2.6 Summary of Current Evidence

Table 2.8 summarizes the principal findings obtained during the exploratory data analysis and their implications for the subsequent preprocessing and model development stages.

Table 2.8: Summary of key findings from the exploratory data analysis.

Domain	**Main Result**	**Implication**
Class balance	$\chi^2(3) = 8110.78$; Cohen's $w = 0.619$	Substantial class imbalance, with the Normal class dominating the dataset and Viral Pneumonia being underrepresented.
Integrity	0 unreadable files; 0 unmatched image–mask pairs	Complete correspondence between chest X-ray images and segmentation masks.
Image encoding	140 RGB X-rays, all belonging to the Viral Pneumonia class	Class-specific technical artifact; all images should be standardized to grayscale before model training.
Exact duplicates	59 redundant files, including 51 COVID-19 images	Duplicate images should be removed prior to dataset partitioning to prevent data leakage.
Lung-mask coverage	$H(3) = 1493.59$; $\varepsilon^2 = 0.0704$	Modest but statistically significant differences in lung-mask coverage across diagnostic classes.
Whole-image brightness	$H(3) = 987.39$; $\varepsilon^2 = 0.0465$	Small but significant class-associated differences in global image brightness.
Whole-image contrast	$H(3) = 1393.78$; $\varepsilon^2 = 0.0657$	Modest but significant class-associated differences in global image contrast.

Overall, the dataset is structurally complete and internally consistent. However, several characteristics were identified that could influence machine learning performance independently of clinically meaningful pathology. These include substantial class imbalance, class-specific RGB image encoding, concentrated exact duplication within the COVID-19 class, statistically significant differences in lung-mask coverage, and systematic differences in whole-image brightness and contrast across diagnostic categories.

Although these characteristics are not inherently problematic, they represent potential technical confounding factors that could encourage a model to learn dataset-specific shortcuts instead of disease-related radiographic features. Consequently, the preprocessing pipeline has been designed to address these issues through deduplication, image standardization, mask alignment, intensity normalization, and stratified dataset partitioning. The effectiveness of these preprocessing steps will be evaluated during model development to ensure that the resulting classifiers learn clinically meaningful representations and generalize reliably to previously unseen data.

Chapter 3

Preprocessing

The exploratory analysis in Chapter 2 identified several characteristics that could bias a classifier independently of pathology: exact duplicate images, class-specific RGB encoding, mismatched image–mask resolutions, class imbalance, and systematic differences in global brightness and contrast. The preprocessing strategy described here addresses these issues before model training while preserving the original dataset and documenting every decision for reproducibility.

3.1 Design Rationale

Each preprocessing choice follows directly from the exploratory findings. Duplicate removal prevents data leakage between training and evaluation sets. Grayscale conversion removes the class-associated RGB encoding found in 10.41% of Viral Pneumonia images. Mask resizing with nearest-neighbor interpolation preserves lung boundaries when aligning 256×256 masks with 299×299 X-rays. Stratified partitioning maintains the imbalanced class distribution across splits. Per-image min–max normalization reduces global intensity differences that may reflect acquisition protocol rather than pathology. Optional contrast enhancement (CLAHE), lung-region masking, and training-time augmentation are retained as controlled alternatives for later comparison.

Figure 3.1 summarizes the overall workflow from the raw collection of 21,165 images to the modeling-ready set of 21,106 samples.

Raw dataset (21,165 images) → remove 59 exact duplicates → cleaned dataset (21,106 images) → stratified split (70% / 15% / 15%) → per-image pipeline (grayscale, resize, optional CLAHE / lung mask / augmentation, min–max normalization) → training, validation, and test partitions

Figure 3.1: End-to-end preprocessing workflow. CLAHE, lung masking, and augmentation were disabled in the baseline configuration but remain available for ablation studies.

3.2 Deduplication and Processed Dataset

Exact duplicate groups identified during exploration yielded 59 redundant files. These were excluded *before* splitting so that identical observations cannot appear in both training and evaluation subsets.

Table 3.1 reports the class distribution after removal. The relative proportions remain close to the raw dataset because duplicates represented only 0.28% of all images and were concentrated in the COVID-19 class, where 51 of 3,616 images (1.41%) were redundant.

Table 3.1: Class distribution after removing 59 exact duplicate files.

Class	**Images**	**Percentage**
COVID-19	3,565	16.89%
Lung Opacity	6,012	28.48%
Normal	10,191	48.29%
Viral Pneumonia	1,338	6.34%
Total	**21,106**	**100.00%**

The 196 image pairs that share a perceptual hash but differ at the pixel level were not removed automatically. They remain flagged for manual or similarity-based review because automatic exclusion could discard legitimately distinct acquisitions.

3.3 Stratified Partitioning

The cleaned dataset was divided using a two-step stratified procedure with a fixed random seed to ensure reproducibility:

1. 70% assigned to training and 30% held out for evaluation.
2. The holdout set divided equally into validation and test subsets, yielding an overall 70% / 15% / 15% partition.

Class proportions were preserved at each step. Table 3.2 lists the resulting sample counts; Table 3.3 shows that normalized class frequencies remain aligned across all three splits, with a maximum deviation of 0.03 percentage points for Viral Pneumonia.

Table 3.2: Absolute sample counts per split after deduplication and stratification.

Class	**Train**	**Validation**	**Test**	**Total**
COVID-19	2,495	535	535	3,565
Lung Opacity	4,208	902	902	6,012
Normal	7,134	1,528	1,529	10,191
Viral Pneumonia	937	201	200	1,338
Total	**14,774**	**3,166**	**3,166**	**21,106**

Table 3.3: Class proportions within each split.

Class	**Train**	**Validation**	**Test**
COVID-19	16.89%	16.90%	16.90%
Lung Opacity	28.48%	28.49%	28.49%
Normal	48.29%	48.26%	48.29%
Viral Pneumonia	6.34%	6.35%	6.32%

3.4 Per-Image Transformation

Each chest X-ray underwent the following sequence of transformations. The order matters: geometric operations precede contrast adjustment and masking, and normalization is applied last so that intensity scaling reflects the final pixel content presented to the model.

1. **Grayscale conversion.** All images were converted to single-channel grayscale, eliminating the RGB encoding bias described in Chapter 2.

2. **Resize.** Images were scaled to 224×224 pixels using area-based interpolation. Segmentation masks were resized to the same dimensions with nearest-neighbor interpolation to preserve binary lung boundaries.

3. **Optional augmentation.** When enabled, geometric and photometric transforms were applied to the image and mask jointly so that spatial correspondence is maintained (Section 3.5).

4. **Optional CLAHE.** Contrast Limited Adaptive Histogram Equalization (clip limit 2.0, tile size 8×8) was applied to the image only, targeting locally varying contrast without altering mask geometry.

5. **Optional lung masking.** Pixels outside the binarized lung region were zeroed before normalization, restricting intensity statistics to pulmonary tissue.

6. **Per-image min–max normalization.** Each image was scaled to $[0, 1]$ using its own minimum and maximum pixel value:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min} + \varepsilon}, \qquad \varepsilon = 10^{-8}.$$

   Independent normalization attenuates scanner- and acquisition-specific global brightness differences identified in the brightness–contrast analysis.

Figure 3.2 illustrates these stages on a representative COVID-19 chest X-ray. The aligned lung mask shows nearest-neighbor resizing from 299×299 to 224×224. CLAHE and lung masking are shown for illustration; both were disabled in the baseline configuration.

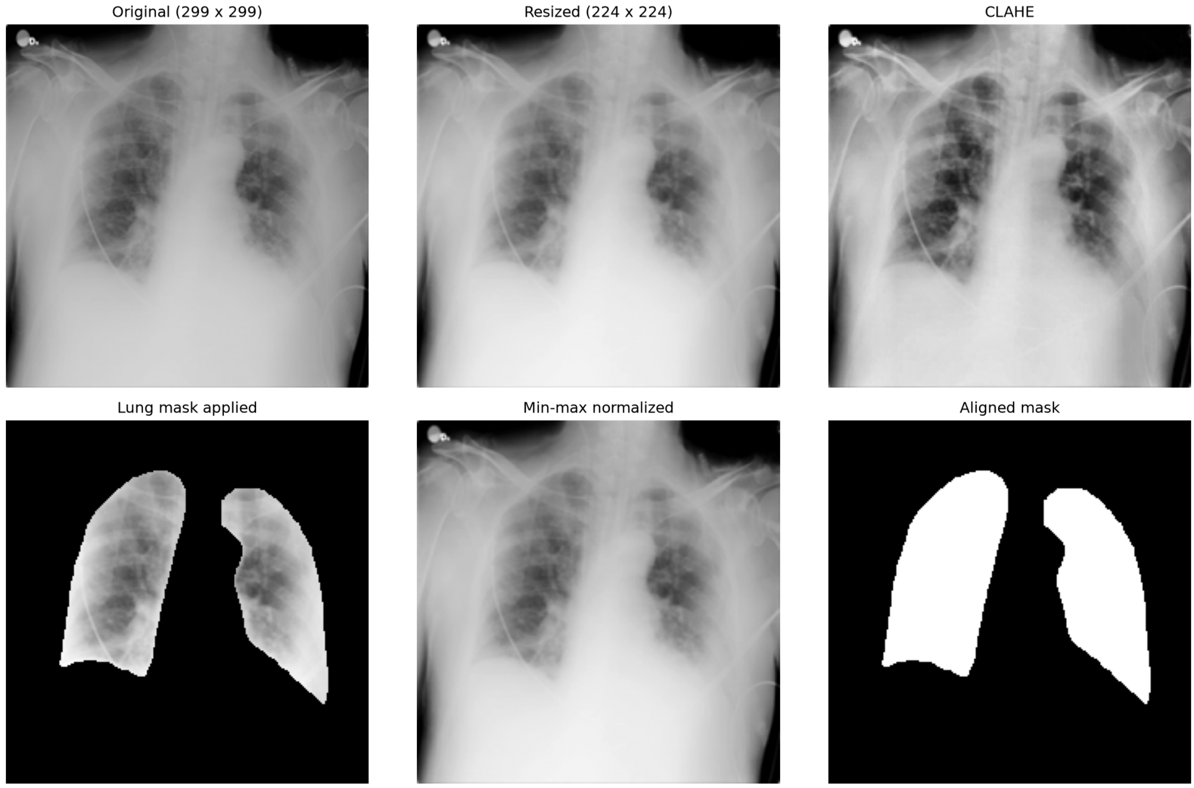


Figure 3.2: Per-image preprocessing stages on a representative COVID-19 chest X-ray: original grayscale image, resize to 224×224, CLAHE, lung masking, min–max normalization, and aligned segmentation mask.

Table 3.4 summarizes the baseline configuration. CLAHE, lung masking, and augmentation were disabled so that their effect on downstream classification can be evaluated as controlled ablations.

Table 3.4: Baseline preprocessing configuration.

Setting	**Baseline value**
Output spatial size	224×224 pixels
Grayscale conversion	Applied to all images
CLAHE	Disabled
Lung mask application	Disabled
Lung mask binarization threshold	127
Training-time augmentation	Disabled
Random seed	42
Train / validation / test ratio	70% / 15% / 15%

3.5 Training-Time Augmentation

Data augmentation was designed to increase effective training-set diversity without breaking the correspondence between chest X-rays and lung segmentation masks. Geometric transforms were applied to both modalities jointly; photometric adjustments affected the image only. Augmentation was restricted to the training partition so that validation and test performance reflect unmodified acquisitions.

Table 3.5 lists the operations and their activation probabilities. Each operation is applied stochastically and independently, so a single training sample may receive none, one, or several transforms.

Table 3.5: Augmentation operations and activation probabilities.

Operation	**Probability**	**Effect**
Horizontal flip	0.5	Mirror image and mask about the vertical axis
Rotation with zoom compensation	0.5	Up to $\pm 15°$; borders reflected on image
Random zoom	0.5	Scale change up to $\pm 8\%$; center crop or pad
Translation	0.5	Shift up to $\pm 5\%$ of width and height
Brightness / contrast	0.4	Intensity scaling up to $\pm 10\%$; image only

Rotation without zoom compensation was evaluated separately and found to introduce empty border regions in the segmentation mask when reflect padding was used. The zoom-compensated rotation variant scales the image to fill the frame and uses zero padding on the mask, keeping lung labels binary. Figure 3.3 compares individual geometric transforms on a representative COVID-19 case. Figure 3.4 shows the combined effect of a single stochastic augmentation pass on the same sample.

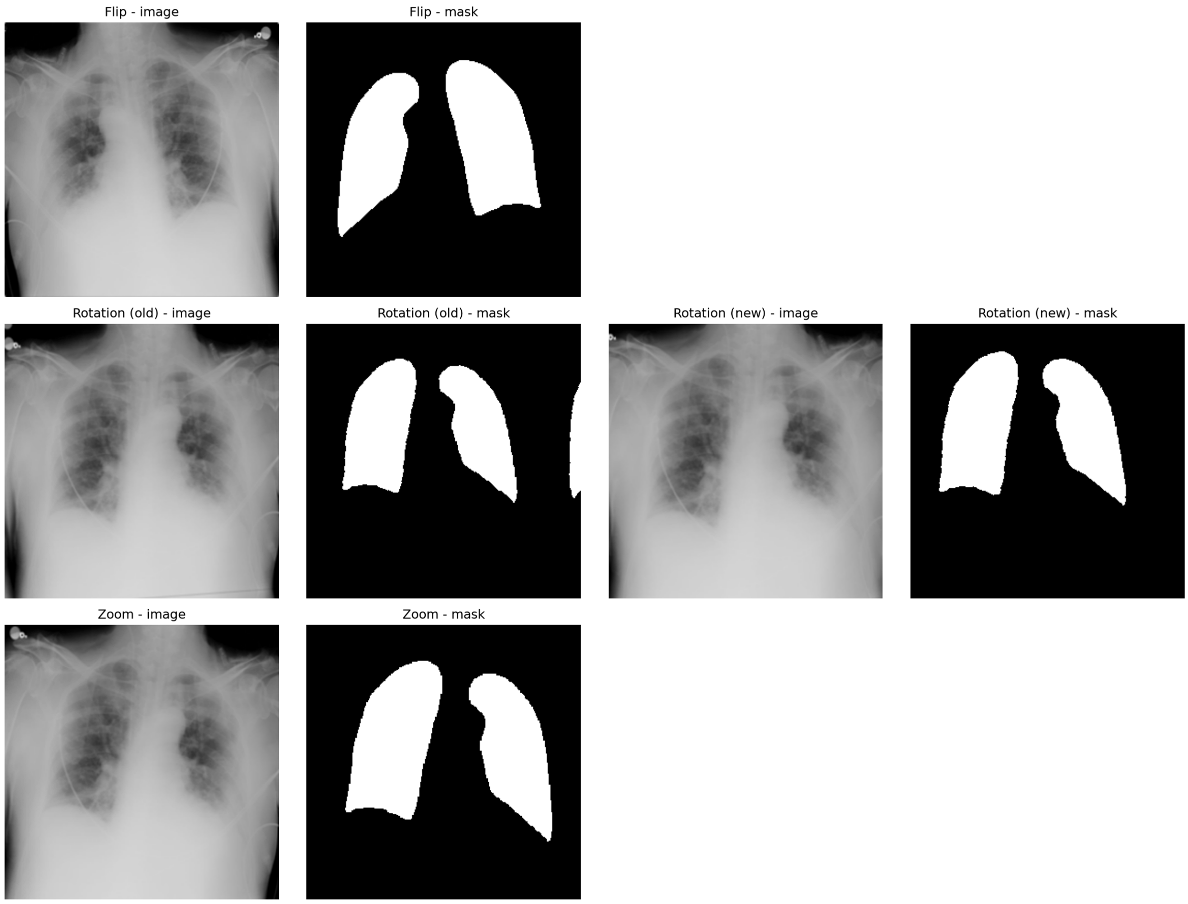


Figure 3.3: Individual augmentation operations on a representative COVID-19 chest X-ray: horizontal flip, rotation without zoom compensation, rotation with zoom compensation, and random zoom. Each row pairs the transformed image with its aligned lung mask.

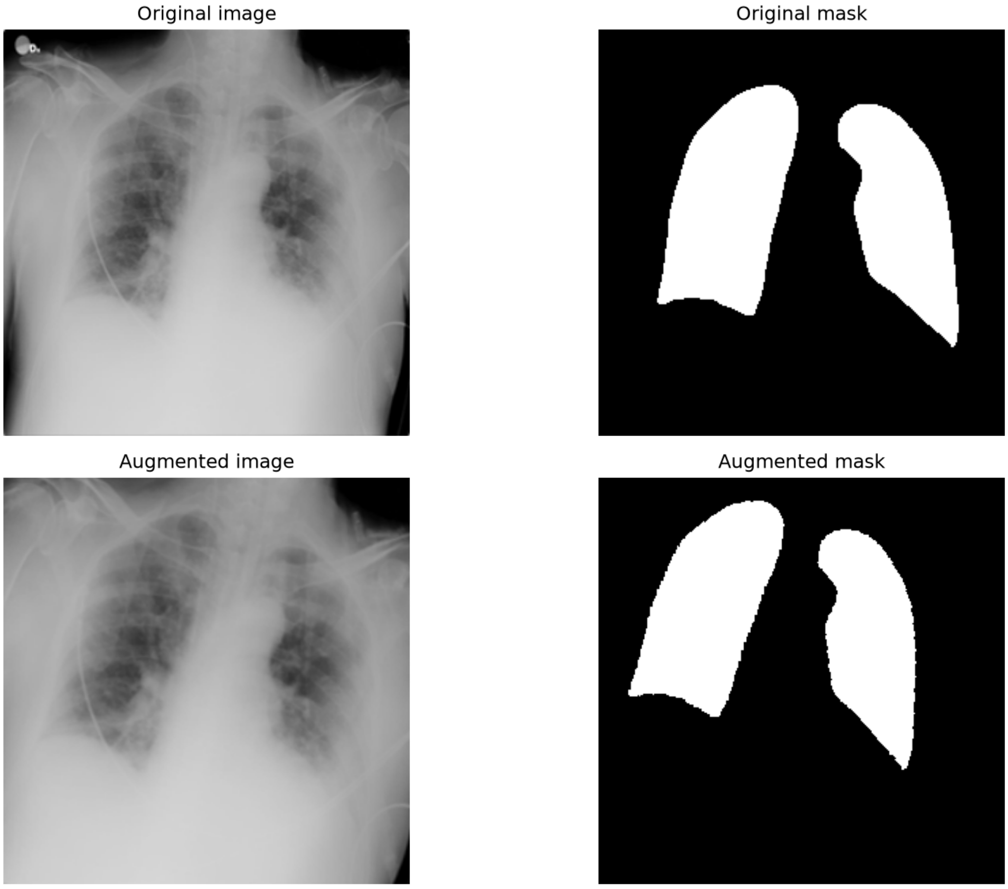


Figure 3.4: Original COVID-19 chest X-ray and lung mask compared with one stochastic augmentation pass (fixed random seed for reproducibility).

3.6 Verification and Reproducibility

The preprocessing workflow was verified at three levels before model training.

First, **transform correctness** was confirmed on synthetic images: normalized outputs lie in $[0, 1]$, repeated processing without augmentation yields identical results, lung masking zeroes non-pulmonary regions, and CLAHE changes pixel values while preserving spatial dimensions. Geometric augmentation preserves image and mask shape across 25 consecutive stochastic draws.

Second, **split integrity** was confirmed: the three partitions are disjoint, together they account for all 21,106 retained samples, and class proportions match the full dataset within 0.05 percentage points.

Third, **end-to-end consistency** was confirmed on the full dataset: all 21,106 images were read successfully with zero failures; processed image counts, split totals, and stacked sample counts all equal 21,106; and a complete rebuild of the baseline configuration produced bit-identical outputs to the saved training, validation, and test partitions (14,774 + 3,166 + 3,166 = 21,106).

A total of 78 automated checks covering transforms, augmentation, splitting, and pipeline regression passed. Stochastic steps used random seed 42 throughout.

3.7 Implications for Model Development

This preprocessing stage establishes a fixed, auditable baseline for convolutional network training. Stratified splitting and duplicate removal reduce leakage risk; grayscale standardization and per-image normalization target the technical shortcuts documented in Chapter 2. Because class imbalance persists after splitting—Viral Pneumonia remains at 6.34% of the training set—downstream evaluation must report per-class precision, recall, macro-averaged F1, balanced accuracy, and confusion matrices rather than overall accuracy alone.

Optional CLAHE, lung masking, and augmentation remain available for systematic comparison in subsequent experiments. Their impact on classification performance and on whether the model attends to pulmonary rather than acquisition-related features will be assessed during model development.

Chapter 4

Baseline Models

Two baseline configurations were established before evaluating more advanced approaches such as data augmentation and transfer learning. The objective was not only to obtain reference classification performance, but also to examine whether the convolutional neural network (CNN) relied on non-pulmonary or dataset-specific visual cues.

The first baseline used the complete chest radiograph and represented an unconstrained classification setting. However, additional region-based experiments showed that substantial predictive information was also present outside the segmented lung fields. Therefore, a second baseline was introduced using a lung-centred region of interest (ROI). In this configuration, the lung segmentation masks were used only to localise the thoracic ROI and were not applied directly to the input image. This avoided the artificial black boundaries introduced by hard lung masking while reducing peripheral image content.

The two baselines therefore served complementary purposes:

- **Full-image baseline:** measures the classification performance achievable from the complete chest X-ray.
- **Lung-ROI baseline:** provides a more spatially controlled reference in which peripheral image regions are reduced while the original radiographic appearance is preserved.

Neither baseline used data augmentation or transfer learning.

4.1 Experimental Setup

Both baselines used the same deliberately simple CNN trained from scratch. The architecture consisted of three convolutional blocks followed by global average pooling and a four-class softmax classifier:

Simple CNN Architecture
Input: $128 \times 128 \times 1$
Conv2D, 16 filters $\rightarrow$ MaxPooling
Conv2D, 32 filters $\rightarrow$ MaxPooling
Conv2D, 64 filters $\rightarrow$ MaxPooling
Global Average Pooling
Dense layer with 4-class softmax output

The model contained 23,556 trainable parameters. The classification task included four classes: COVID, Lung Opacity, Normal, and Viral Pneumonia. The same deduplicated dataset, train/validation/test partitions, random seed, class-weighting strategy, and optimisation settings were maintained across both baseline experiments.

The class distributions were identical for the two baselines and are shown in Table 4.1.

Table 4.1: Class distribution used for the baseline experiments.

Class	**Train**	**Validation**	**Test**
COVID	16.89%	16.90%	16.90%
Lung Opacity	28.48%	28.49%	28.49%
Normal	48.29%	48.26%	48.29%
Viral Pneumonia	6.34%	6.35%	6.32%

Class weighting was used to reduce the effect of class imbalance. Training was performed for a maximum of 100 epochs with early stopping based on validation loss.

4.2 Baseline 1: Full-Image Simple CNN

The first baseline was trained directly on the complete chest radiograph, without augmentation or additional intensity or geometry normalisation.

Its test-set performance is presented in Table 4.2.

Table 4.2: Test-set performance of the full-image simple CNN baseline.

Class	**Precision**	**Recall**	**F1-score**
COVID	0.766	0.830	0.796
Lung Opacity	0.799	0.766	0.782
Normal	0.850	0.831	0.840
Viral Pneumonia	0.840	0.945	0.889
Macro average	0.813	0.843	**0.827**

The full-image model achieved a test accuracy of **81.96%** and a macro F1-score of **82.7%**. This represents the strongest raw baseline performance obtained with the simple CNN.

However, the full-image model should be interpreted as a *naive performance baseline*. A separate background-only experiment achieved approximately 73.40% test accuracy, indicating that non-lung regions contain substantial class-correlated information. Therefore, high full-image classification performance does not necessarily imply that the model relies exclusively on clinically relevant pulmonary features.

4.2.1 Grad-CAM Analysis of the Full-Image Baseline

Grad-CAM was applied to the full-image model to investigate the spatial regions associated with its predictions. Quantitative attention analysis was performed on the complete test set of 3,166 images.

The mean lung attention fraction was 20.7%, whereas the segmented lungs occupied approximately 24.0% of the image area. The resulting mean lung-attention enrichment was 0.868. An enrichment value of 1 indicates that attention is approximately proportional to lung area, whereas values below 1 indicate relative under-attention to the lung fields.

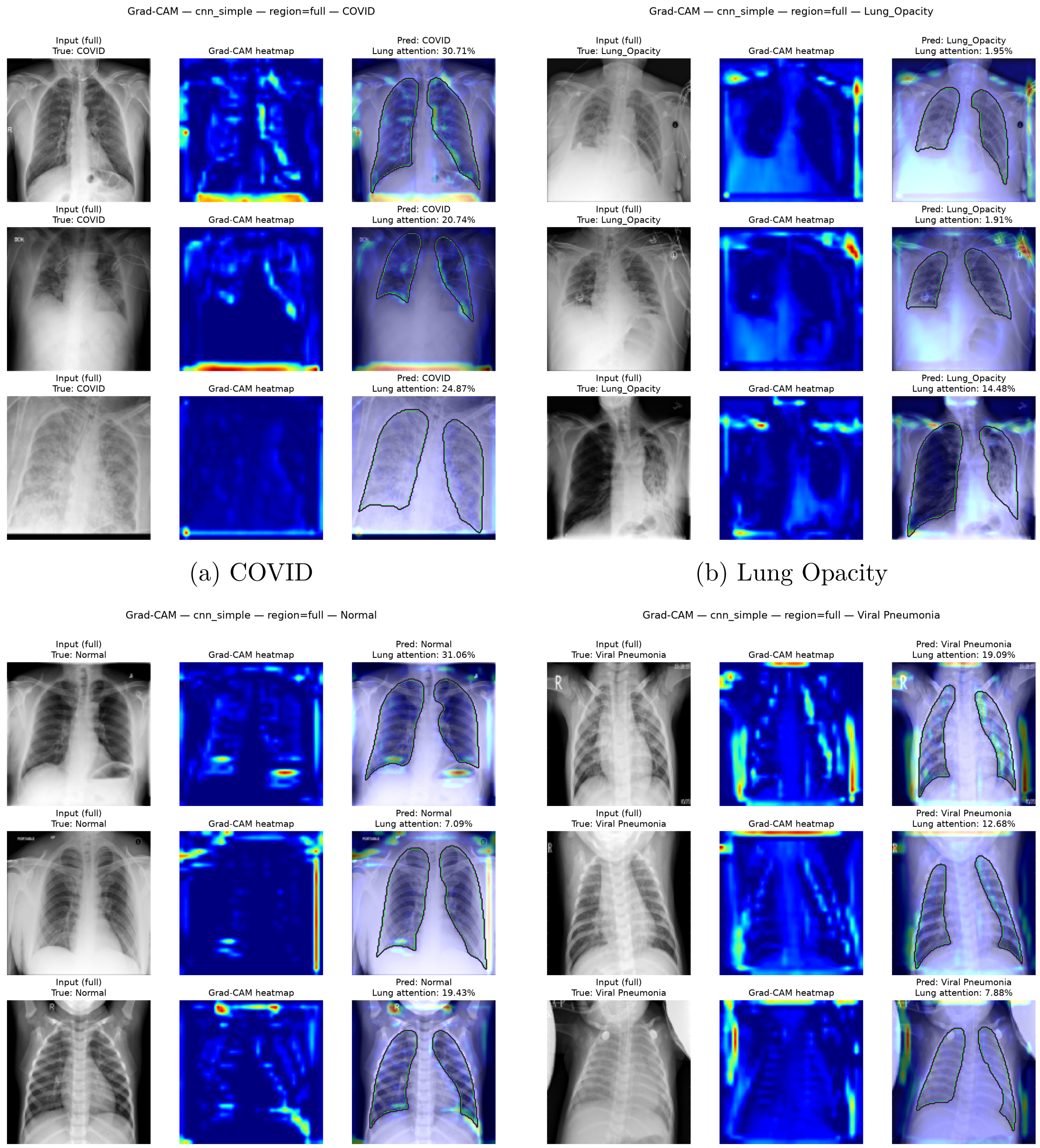
Class-specific results are reported in Table 4.3.

Table 4.3: Class-wise Grad-CAM lung attention for the full-image baseline.

Class	**Lung Attention**	**Lung Area**	**Enrichment**
COVID	24.46%	24.38%	1.063
Lung Opacity	11.63%	21.80%	0.518
Normal	25.24%	25.18%	1.024
Viral Pneumonia	16.81%	23.22%	0.731

The results indicate strongly class-dependent attention behaviour. COVID and Normal cases exhibited approximately proportional lung attention, whereas Lung Opacity and Viral Pneumonia showed substantially lower lung-attention enrichment.

Qualitative inspection further revealed heterogeneous attention patterns. Some predictions were associated with activations within the pulmonary fields, while others showed strong responses near image borders or outside the segmented lung regions. These findings suggest that the full-image CNN may use a combination of pulmonary information and non-pulmonary or dataset-specific visual cues.



(a) COVID

(b) Lung Opacity

(c) Normal

(d) Viral Pneumonia

Figure 4.1: Representative Grad-CAM examples for the full lung baseline, grouped by class.

4.2.2 Motivation for a Lung-Centred ROI Baseline

A hard lung-masking experiment was initially used to restrict the CNN to the segmented pulmonary regions. This model achieved 72.93% test accuracy and 69.0% macro F1. However, hard masking introduced a sharp artificial boundary between the retained lung pixels and the zero-valued background.

Grad-CAM visualisations frequently showed activation around these segmentation boundaries. Consequently, the reduction in performance could not be attributed solely to the removal of background information, because the masking operation itself changed the image representation and potentially introduced new geometric cues.

To obtain a cleaner controlled baseline, a lung-centred ROI preprocessing strategy was therefore introduced.

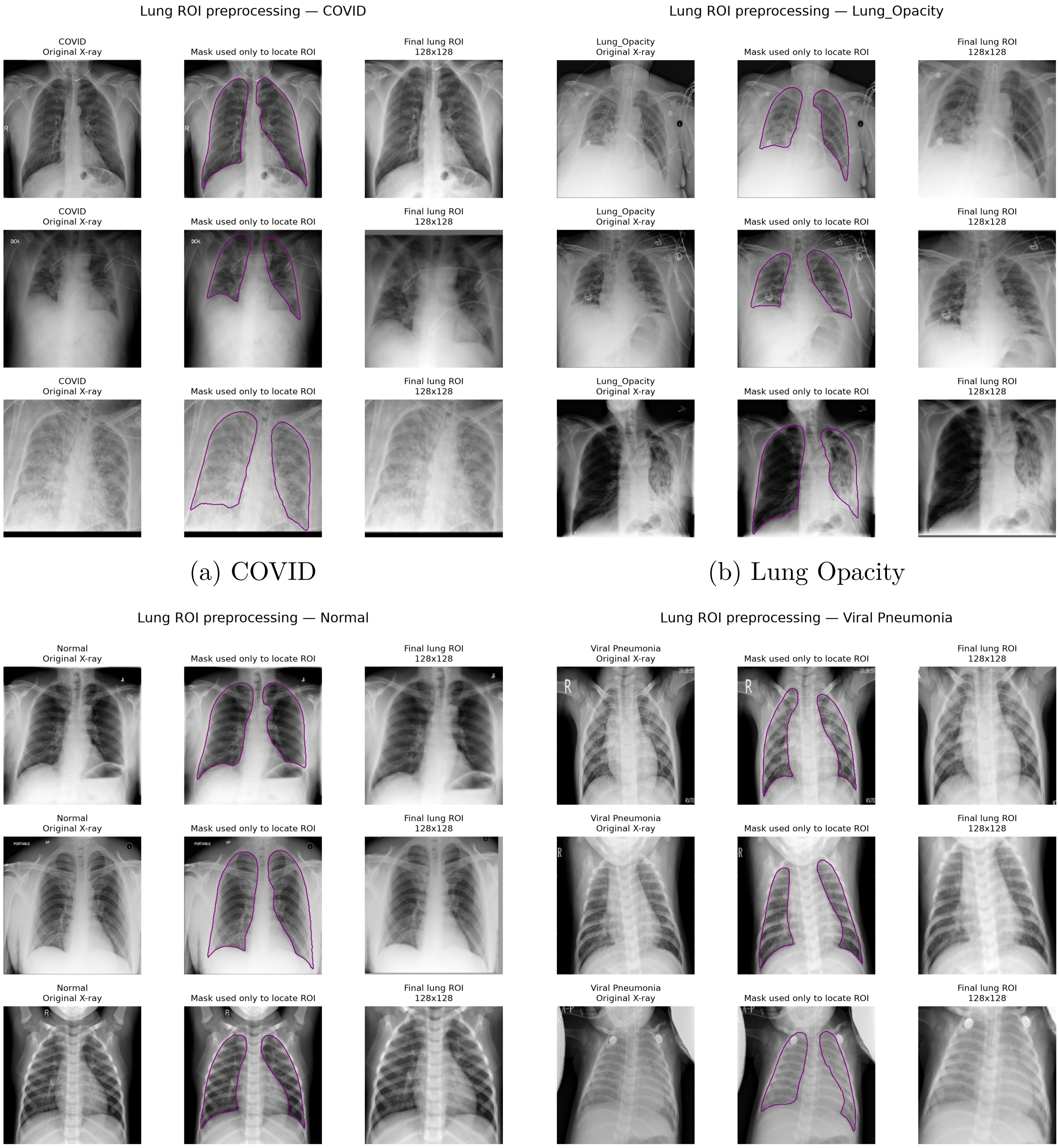
4.3 Baseline 2: Lung-ROI Simple CNN

For the second baseline, lung segmentation masks were used only to determine the spatial extent of the lungs. The image itself was not multiplied by the mask.

The preprocessing pipeline was:

1. Load the original chest X-ray and corresponding lung mask.
2. Use the mask to determine the joint bounding box of the two lungs.
3. Add a small spatial margin around the lung region.
4. Construct a square ROI around the lung pair.
5. Preserve the original radiographic intensities and anatomical appearance.
6. Resize the ROI to 128×128 pixels.

This preprocessing reduced variation in overall framing and lung scale while avoiding hard segmentation boundaries. No rotation normalisation, intensity normalisation, or augmentation was applied.



(a) COVID (b) Lung Opacity

(c) Normal (d) Viral Pneumonia

Figure 4.2: Class-wise examples of the lung-centred ROI preprocessing pipeline.

The lung-ROI model achieved the test-set performance shown in Table 4.4.

Table 4.4: Test-set performance of the lung-ROI simple CNN baseline.

Class	Precision	Recall	F1-score
COVID	0.649	0.703	0.675
Lung Opacity	0.822	0.691	0.751
Normal	0.828	0.853	0.840
Viral Pneumonia	0.746	0.940	0.832
Macro average	0.761	0.797	**0.774**

The lung-ROI model achieved **78.71% test accuracy** and **77.4% macro F1**. Its train, validation, and test accuracies were 79.86%, 78.74%, and 78.71%, respectively, indicating close agreement between the data partitions and little evidence of severe overfitting.

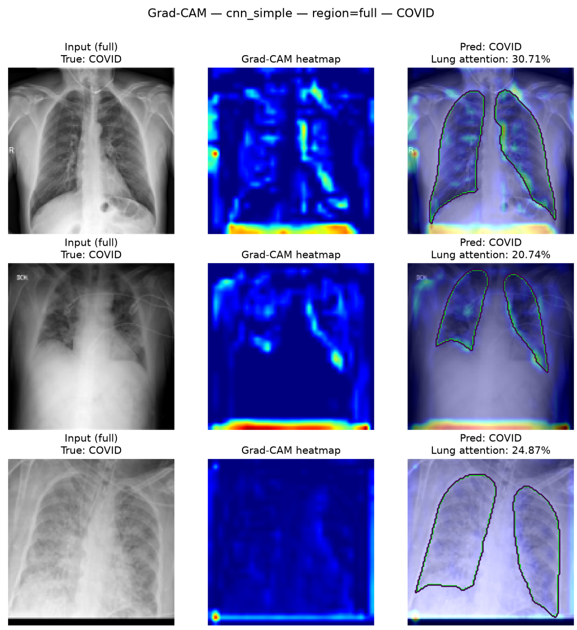
4.3.1 Grad-CAM Analysis of the Lung-ROI Baseline

Grad-CAM was also applied to the lung-ROI baseline. Unlike the hard-masked experiment, no artificial segmentation contour was present in the CNN input. Therefore, activation outside the segmented lung contour corresponds to actual anatomical or image regions contained within the ROI.

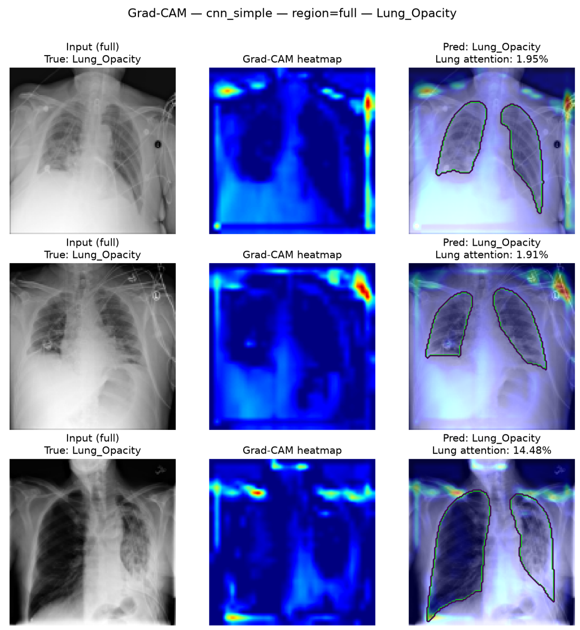
Qualitative inspection showed heterogeneous attention across the four classes. COVID examples generally demonstrated a greater degree of pulmonary activation, although attention was not exclusively confined to the lung fields. In several Lung Opacity and Viral Pneumonia examples, correctly classified images exhibited strong activation in peripheral, inferior, mediastinal, or ROI-border regions despite relatively low activation within the segmented pulmonary interior. Normal cases showed mixed behaviour, with both pulmonary and peripheral activation patterns.

These observations suggest that ROI cropping substantially reduces peripheral image information but does not completely prevent the CNN from exploiting extra-pulmonary or framing-related cues within the cropped thoracic region.

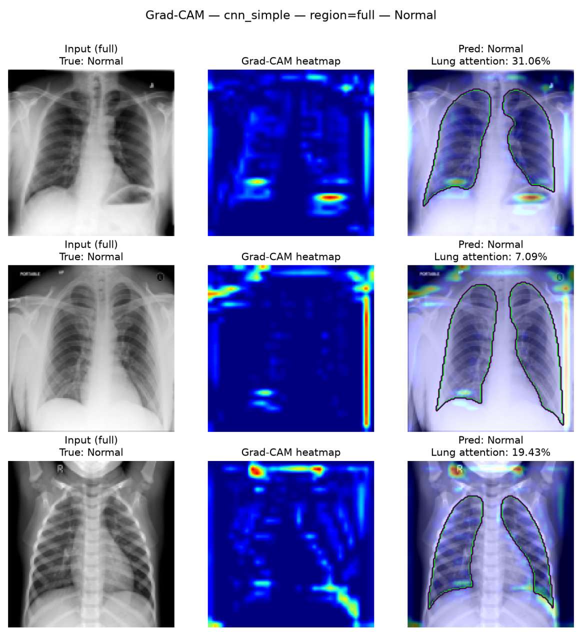
Because this ROI Grad-CAM assessment is currently based on representative qualitative examples rather than complete test-set quantitative attention analysis, the findings are interpreted as exploratory.



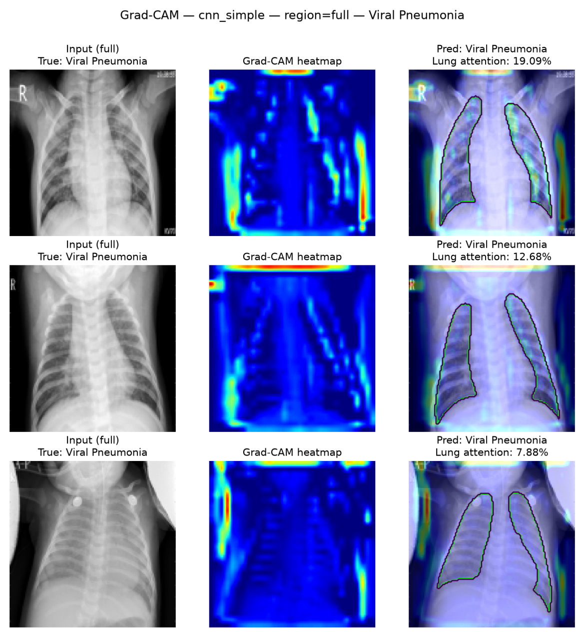
(a) COVID



(b) Lung Opacity



(c) Normal



(d) Viral Pneumonia

Figure 4.3: Representative Grad-CAM visualisations for the lung-ROI baseline, grouped by class. Although the ROI reduces peripheral image content and avoids hard segmentation boundaries, model attention remains heterogeneous and is not restricted exclusively to the pulmonary parenchyma.

4.3.2 Comparison of the Two Baselines

Table 4.5 compares the two principal baselines.

Table 4.5: Comparison of the two principal simple-CNN baselines.

Baseline	Test Accuracy	Macro F1
Full-image simple CNN	**81.96%**	**82.7%**
Lung-ROI simple CNN	78.71%	77.4%

Restricting the model to the lung-centred ROI reduced test accuracy by 3.25 percentage points and macro F1 by 5.3 percentage points. Nevertheless, the ROI model retained a substantial proportion of the full-image classification performance.

The class-wise recall comparison is shown in Table 4.6.

Table 4.6: Class-wise recall of the full-image and lung-ROI baselines.

Class	Full Image	Lung ROI
COVID	**83.0%**	70.3%
Lung Opacity	**76.6%**	69.1%
Normal	83.1%	**85.3%**
Viral Pneumonia	**94.5%**	94.0%

The largest reduction occurred for COVID classification. In contrast, Normal and Viral Pneumonia recall remained relatively stable.

These results suggest that peripheral image regions contribute to the performance of the full-image model, but they are not the sole source of discriminative information. A substantial level of classification performance remained when the input was restricted to a lung-centred thoracic ROI.

4.4 Baseline Interpretation and Role in Subsequent Experiments

The two baselines provide complementary reference points for subsequent model development.

The full-image model achieved the highest baseline performance, with 81.96% test accuracy and 82.7% macro F1. However, the background-only experiment and Grad-CAM analysis indicate that non-lung regions contain substantial class-correlated information. Consequently, the full-image model is retained as an unconstrained or naive performance baseline.

The lung-ROI model achieved 78.71% test accuracy and 77.4% macro F1 while substantially reducing peripheral image content and avoiding the hard-mask boundary artefact. It therefore serves as a more spatially controlled second baseline.

These configurations will be retained when evaluating subsequent augmentation and transfer-learning approaches.

This design supports two complementary questions:

1. Does a more advanced model improve overall classification performance on the complete chest radiograph?

2. Does the model also improve performance when access to obvious peripheral and background cues is reduced?

The second comparison is particularly important because an improvement in full-image accuracy alone does not necessarily imply improved use of clinically relevant pulmonary information.

Chapter 5

Transfer Learning

5.1 Transfer-Learning Approach and Experimental Setup

In addition to the two simple CNN baselines trained from scratch (Section 4), a transfer-learning approach was evaluated using an EfficientNetB0 backbone pretrained on ImageNet. The motivation was to test whether features learned on a large, general-purpose image dataset could improve classification performance on a relatively small, domain-specific chest radiograph dataset, and whether such a model relies on the same visual cues as the CNN trained from scratch.

The classification head placed on top of the backbone consisted of:

Transfer-Learning Head
EfficientNetB0 backbone (ImageNet weights)
Global Average Pooling
Dropout (rate 0.3)
Dense layer, 128 units, ReLU
Dropout (rate 0.3)
Dense layer, 4-class softmax output

Training used the same manifest, the same 70/15/15 stratified train, validation, and test split, and the same random seed (42) as the CNN baselines, so results are directly comparable to those reported in Section 4.

A two-phase training design was used throughout this chapter. In phase 1, the EfficientNetB0 backbone was kept frozen and only the classification head was trained. In several experiments (Section 5.4), a second phase then unfroze the backbone and continued training end-to-end at a substantially lower learning rate. This design allowed the new head to first adapt to the pretrained features before any backbone weights were disturbed, reducing the risk of destructive early updates.

5.2 Baseline Transfer Model

The first transfer-learning experiment kept the EfficientNetB0 backbone entirely frozen and trained only the classification head, without augmentation or class weighting. Its test-set performance is presented in Table 5.1.

Table 5.1: Test-set performance of the frozen EfficientNetB0 transfer model.

Class	Precision	Recall	F1-score
COVID	0.943	0.903	0.923
Lung Opacity	0.906	0.808	0.854
Normal	0.869	0.946	0.906
Viral Pneumonia	0.984	0.910	0.945
Macro average	0.925	0.892	**0.907**

The frozen transfer model achieved a test accuracy of **89.70%** and a macro F1-score of **90.7%**. This already exceeds the strongest simple-CNN baseline (the full-image model at 81.96% accuracy and 82.7% macro F1, Section 4.2) by approximately 7.7 accuracy points and 8.0 macro-F1 points, without any fine-tuning of the pretrained backbone. This result confirms that ImageNet-derived features transfer usefully to this chest radiograph classification task, even though the source and target domains are visually very different.

5.3 Optimisation: Augmentation, Class Weighting, and the Generalisation Gap

Following the initial baseline, several optimisation experiments were conducted to evaluate whether standard machine-learning tools for improving generalisation and handling class imbalance could improve on the frozen baseline, in line with the project's Step 2 objective of using statistics and optimisation tools to fully exploit and understand the model's results.

5.3.1 Data Augmentation and the Effect of Horizontal Flipping

An augmentation pipeline combining random horizontal flipping and small random rotations was first applied during training. Chest radiographs are not perfectly left-right symmetric in a clinically meaningless way, since heart position and other incidental laterality cues can appear in the images, so a second variant disabled horizontal flipping and used rotation only. Table 5.2 compares the two variants on the test split.

Table 5.2: Effect of horizontal flipping on the augmented transfer model (test split).

Model	Accuracy	Macro F1	COVID F1	Lung Opacity F1	Normal F1	Viral Pneumonia F1
Flip + rotation	87.62%	0.884	0.879	0.832	0.892	0.935
Rotation only	**88.38%**	**0.896**	0.883	**0.850**	0.895	**0.956**

Removing horizontal flipping improved every test-set metric, with the largest gains on Lung Opacity (+1.85 F1 points) and Viral Pneumonia (+2.07 F1 points). This is consistent with horizontal flipping introducing a mild, unhelpful distribution shift for this task rather than a useful invariance, and suggests that flipping should be treated as a tunable hyperparameter for chest radiographs rather than an unquestioned default.

5.3.2 Class Weighting

A separate experiment applied class weights inversely proportional to class frequency, without augmentation, to counter the dataset's imbalance between the majority Normal and Lung Opacity classes and the minority COVID and Viral Pneumonia classes. The class-weighted model reached **88.34%** test accuracy and **0.894** macro F1, close to the augmented variants. Class weighting mainly redistributed performance toward the minority classes rather than closing the gap between training and test performance, indicating that it addresses a different problem (class imbalance) than augmentation (overfitting).

5.3.3 Generalisation Gap Across Runs

Table 5.3 reports the gap between train and test performance for the frozen baseline and its two augmentation variants, as a simple diagnostic of overfitting.

Table 5.3: Train-test generalisation gap for the frozen-backbone transfer models.

Model	Test Accuracy	Accuracy Gap	Macro-F1 Gap
Frozen baseline (no augmentation)	89.70%	2.77 pts	2.67 pts
Flip + rotation augmentation	87.62%	2.72 pts	2.66 pts
Rotation-only augmentation	88.38%	**1.98 pts**	**1.27 pts**

Rotation-only augmentation roughly halves the frozen baseline's generalisation gap (1.98 versus 2.77 accuracy points, 1.27 versus 2.67 macro-F1 points) for a comparatively small reduction in raw test accuracy. This represents a genuine optimisation trade-off: a model that generalises somewhat better within this dataset, at a small cost in headline performance, although it does not on its own resolve the background-reliance concerns raised in Section 5.5.

5.4 Advanced Modelling: Fine-Tuning the Backbone

Following the project's Step 3 objective of advanced modelling using deep learning, the EfficientNetB0 backbone was unfrozen and fine-tuned end-to-end after the initial frozen-head phase. Phase 1 trained the head only for 10 epochs; phase 2 then unfroze the entire backbone and continued training for 15 further epochs at a much lower learning rate (10^{-5}, versus 10^{-3} for the head-only phase), for 25 epochs in total.

Figure 5.1 shows the resulting training history. Validation loss briefly increases at the start of phase 2, a normal consequence of unfreezing a large pretrained backbone and updating it for the first time, before decreasing steadily to a final value substantially lower than at the end of phase 1 (from 0.310 to 0.189), with validation accuracy rising from 88.9% to 93.3% over the fine-tuning phase.

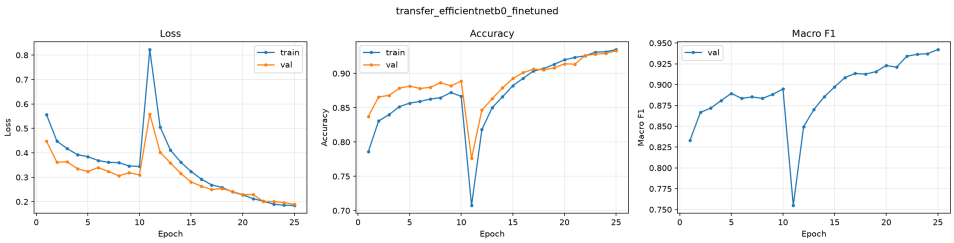


Figure 5.1: Training and validation curves for the fine-tuned EfficientNetB0 model, showing the frozen-head phase followed by end-to-end fine-tuning.

The fine-tuned model's test-set performance is presented in Table 5.4.

Table 5.4: Test-set performance of the fine-tuned EfficientNetB0 transfer model.

Class	**Precision**	**Recall**	**F1-score**
COVID	0.931	0.978	0.954
Lung Opacity	0.958	0.814	0.880
Normal	0.898	0.962	0.929
Viral Pneumonia	0.980	0.980	0.980
Macro average	0.942	0.933	**0.936**

The fine-tuned model achieved a test accuracy of **92.36%** and a macro F1-score of **93.6%**, the strongest result obtained anywhere in this project. Fine-tuning improved every class relative to the frozen baseline except Lung Opacity, whose F1-score changed only marginally (0.854 to 0.880), and produced a particularly large gain in COVID recall (90.3% to 97.8%). The confusion matrix for this model is shown in Figure 5.2.

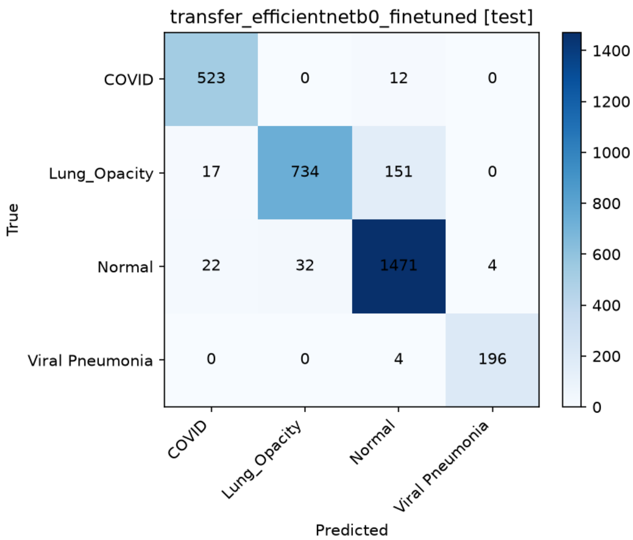


Figure 5.2: Test-set confusion matrix for the fine-tuned EfficientNetB0 model.

As with the full-image simple-CNN baseline, however, this headline performance must be interpreted alongside the interpretability analysis in Section 5.5 before it is read as evidence of genuinely improved pulmonary diagnosis.

5.5 Interpretability: Grad-CAM and the Lung-Focus Causal Analysis

Grad-CAM was applied to the transfer-learning models using the same lung segmentation masks employed for the CNN baselines' interpretability analysis (Section 4.2.1). Beyond qualitative visualisation, a quantitative *lung-focus* metric was computed for every image: the share of each Grad-CAM heatmap's total activation that falls inside the segmented lung mask. This is compared against a *chance* level, defined as the lung mask's own share of the image area, that is, the score a spatially uninformative model would obtain by construction. A model that attends to the lungs no more than chance provides no evidence that it is using pulmonary information specifically.

5.5.1 Qualitative and Quantitative Lung Focus of the Frozen Model

Figure 5.3 shows representative Grad-CAM examples for the frozen transfer model, with the lung boundary overlaid in green. Heatmaps frequently activate at chest borders, shoulders, or corners rather than tightly within the lung fields.

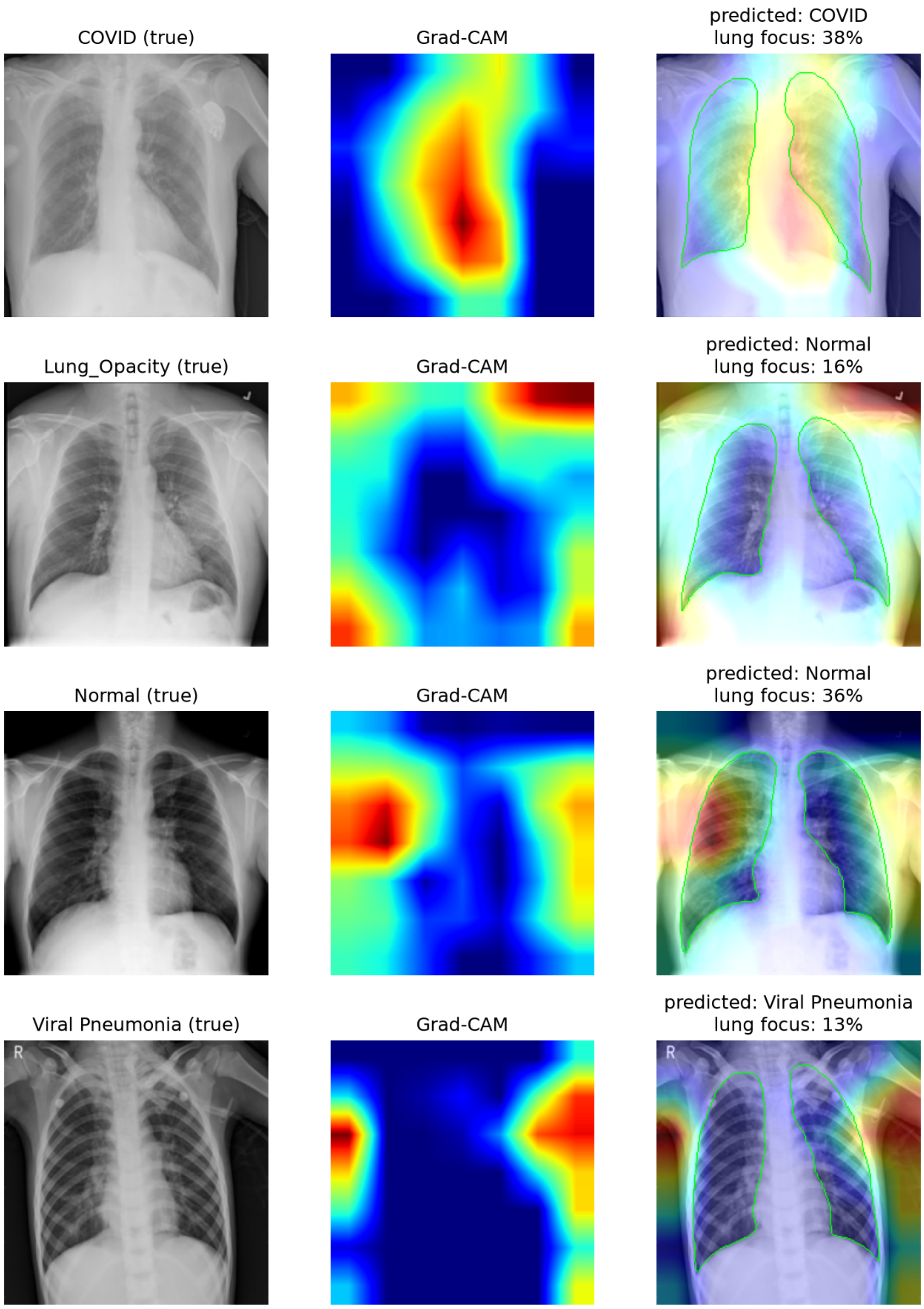


Figure 5.3: Representative Grad-CAM examples for the frozen transfer model, with the lung mask boundary overlaid in green.

Table 5.5 quantifies this over 300 sampled test images.

Table 5.5: Grad-CAM lung focus versus chance for the frozen transfer model.

Class	Grad-CAM in Lungs	Chance Level	Difference
COVID	30.8%	24.7%	+6.1 pts
Normal	28.5%	25.0%	+3.5 pts
Viral Pneumonia	29.8%	25.7%	+4.2 pts
Lung Opacity	17.7%	20.6%	−2.9 pts

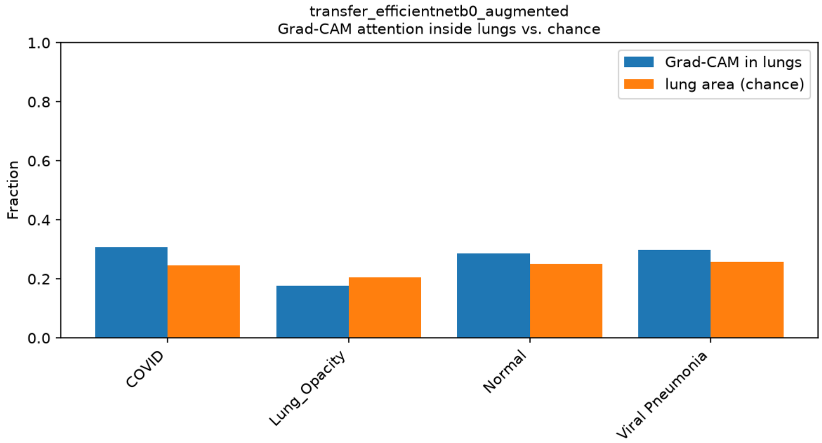


Figure 5.4: Grad-CAM attention inside the lungs versus the chance level, by class, for the frozen transfer model.

Attention is only mildly above chance for three of the four classes and below chance for Lung Opacity, meaning the model attends to non-lung regions more than pure geometry would predict for that class. Splitting the same sample by prediction outcome showed virtually identical lung focus for correct predictions (25.4% versus a chance level of 23.7%) and misclassified predictions (25.2% versus 22.5%), indicating that weak lung-focused attention is a systemic property of this model rather than an error-specific artefact.

5.5.2 Causal Test: Masking the Input

Correlational Grad-CAM evidence cannot by itself establish that the model's accuracy depends on non-pulmonary cues. A direct causal test was therefore performed by training an otherwise identical model on images with every pixel outside the lung mask forcibly set to zero before reaching the network, so that no background information was available at all.

Table 5.6 compares this masked model against the frozen baseline.

Table 5.6: Effect of masking the background at the input (causal test), test split.

Metric	Full Image	Lungs Only (Masked)	Change
Test accuracy	89.70%	83.07%	−6.6 pts
Test macro F1	0.907	0.823	−8.4 pts
COVID recall	90.3%	**57.0%**	−33.3 pts
COVID F1	0.923	0.678	−24.5 pts
Lung Opacity F1	0.854	0.809	−4.5 pts
Normal F1	0.906	0.872	−3.4 pts
Viral Pneumonia F1	0.945	0.932	−1.3 pts

If the model's accuracy depended evenly on pulmonary pathology, removing the background should have degraded every class by a comparable amount. Instead, the loss is heavily concentrated in COVID: recall collapses from 90.3% to 57.0%, a 33.3-point drop, meaning the masked model misses more than 40% of COVID cases that the full-image model correctly identified. The other three classes lose only a few points each. This pattern is direct causal evidence of COVID-specific background leakage in this dataset, consistent with the widely reported concern that COVID images in this Kaggle collection were aggregated from different sources than the other classes, introducing incidental correlations unrelated to disease status.

As expected, Grad-CAM attention moves substantially into the lungs once the background is physically removed, since no other information remains available to the network (Table 5.7, Figure 5.5).

Table 5.7: Grad-CAM lung focus versus chance for the masked transfer model.

Class	Grad-CAM in Lungs	Chance Level	Difference
COVID	40.8%	24.7%	+16.1 pts
Lung Opacity	30.2%	20.6%	+9.6 pts
Normal	49.4%	25.0%	+24.4 pts
Viral Pneumonia	44.4%	25.7%	+18.7 pts

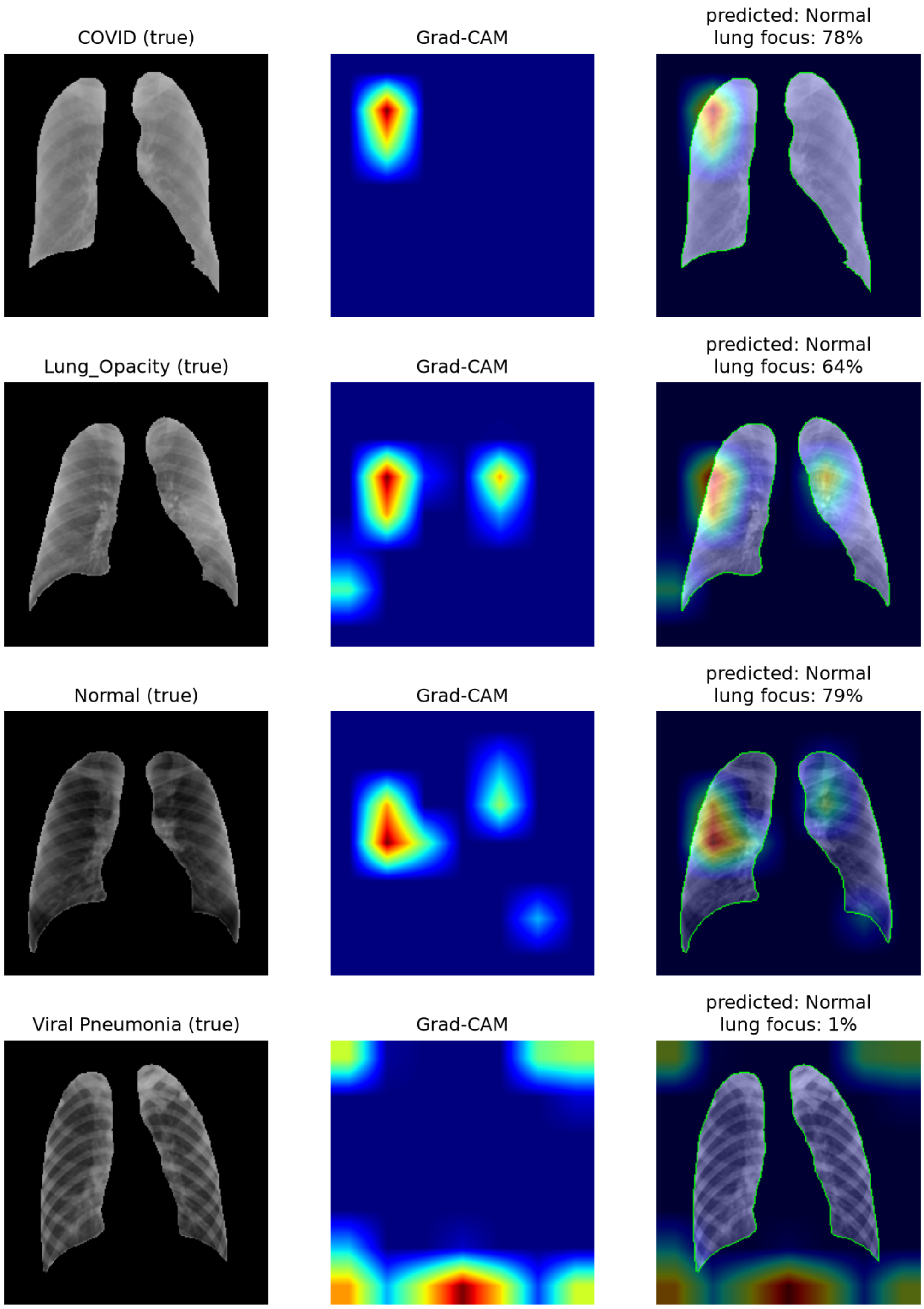


Figure 5.5: Representative Grad-CAM examples for the masked transfer model, with the lung mask boundary overlaid in green. All model input outside the boundary was zeroed before training and inference.

A small number of heatmap regions still extend outside the lung boundary even though the corresponding input pixels were exactly zero. This is not evidence that the masking failed; it is a known limitation of Grad-CAM on deep convolutional networks, caused by non-zero learned biases and batch normalisation statistics propagating through many stacked layers, together with the coarse spatial resolution of the final convolutional feature map and its associated large receptive field. For this reason, the aggregated lung-focus statistics, and above all the masked-training accuracy experiment itself, are treated as the primary evidence, rather than any single Grad-CAM pixel.

5.5.3 Isolating Shape: A Mask-Only Model

The masked model in Section 5.5.2 still receives the lung's pixel intensities, that is, its texture and density patterns, and only removes the background. To isolate lung *shape* from lung *texture*, a further model was trained on the segmentation masks themselves, that is, a binary lung silhouette with no pixel-intensity information from the original radiograph at all.

Table 5.8 extends the comparison to this shape-only model.

Table 5.8: Decomposing accuracy into background, texture, and shape contributions (test split).

Metric	Full Image	Lungs, Texture Kept	Shape Only
Test accuracy	89.70%	83.07%	**74.07%**
Test macro F1	0.907	0.823	**0.703**
COVID recall	90.3%	57.0%	**29.3%**
COVID F1	0.923	0.678	0.413
Normal F1	0.906	0.872	0.818
Lung Opacity F1	0.854	0.809	0.709
Viral Pneumonia F1	0.945	0.932	0.873

Lung shape alone is a weak classifier for COVID, with recall of 29.3%, close to what a random four-class guess would achieve, while Normal cases remain comparatively easier to recognise from shape alone (F1 0.818). The test confusion matrix (Figure 5.6) shows that, of the true COVID cases, the shape-only model most often mistakes them for Normal or Lung Opacity, consistent with COVID's radiographic signature being primarily about tissue density and opacity patterns inside the lung field rather than its outline.

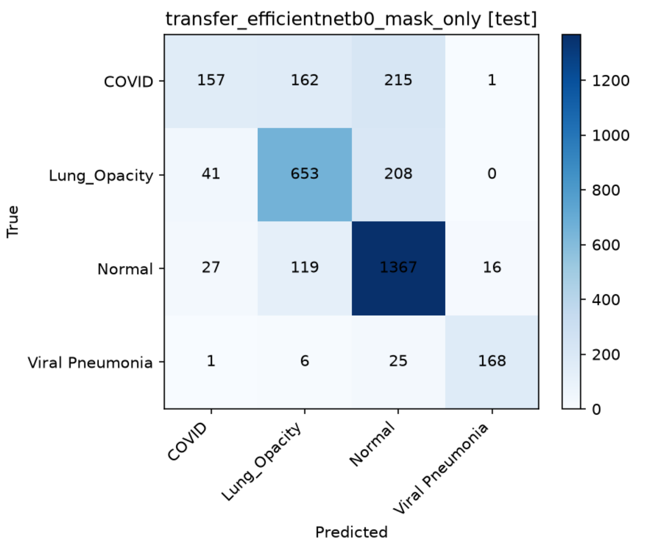


Figure 5.6: Test-set confusion matrix for the shape-only (mask-only) transfer model.

Taken together, Sections 5.5.2 and 5.5.3 give a three-way decomposition of the frozen model's 89.70% test accuracy: approximately 6.6 points depend on non-lung background and context, a further approximately 9.0 points depend on lung texture and density patterns rather than shape, and the remainder is what lung geometry alone can achieve, well above the 25% random baseline but a weak classifier by itself, and specifically unreliable for COVID.

5.5.4 An Architectural Alternative: Masked Pooling

Hard input masking (Section 5.5.2) is an effective causal probe, but it discards all information outside the lungs even where that information might legitimately help, for example by providing context for image quality or overall framing, and it costs several points of raw performance. As a softer alternative, a masked global average pooling layer was used instead: rather than zeroing the input image, the model's final convolutional feature map is masked so that only lung-covering spatial locations contribute to the pooled features that feed the classification head, while the backbone still observes the full image.

This configuration, combined with class balancing and rotation-only augmentation, reached **88.79%** test accuracy and **0.897** macro F1, recovering to within roughly one accuracy point of the frozen full-image baseline while still architecturally constraining the features used for classification to the lung region. This suggests that how a model is restricted to the lungs, that is, at the input versus within the pooling operation, materially affects the performance cost of enforcing that constraint, and is a useful direction for future work aiming to reduce background reliance without the full accuracy cost of hard input masking.

5.5.5 Grad-CAM on the Fine-Tuned Model

The same lung-focus analysis was repeated for the fine-tuned model of Section 5.4, the best-performing model in this project. Table 5.9 reports the per-class results.

Table 5.9: Grad-CAM lung focus versus chance for the fine-tuned transfer model.

Class	**Grad-CAM in Lungs**	**Chance Level**	**Difference**
COVID	24.3%	24.4%	−0.1 pts
Lung Opacity	31.2%	20.9%	+10.3 pts
Normal	38.5%	25.2%	+13.3 pts
Viral Pneumonia	23.8%	24.0%	−0.2 pts

Fine-tuning changes which classes show above-chance lung attention: Lung Opacity and Normal now show clearly above-chance attention, the opposite of the frozen baseline's pattern in Table 5.5, while COVID and Viral Pneumonia sit almost exactly at chance. This differs from the CNN-from- scratch full-image baseline's attention pattern (Section 4.2.1), where COVID and Normal were close to proportional and Lung Opacity was the weakest. The two architectures therefore appear to rely on different, model-specific combinations of pulmonary and non-pulmonary cues, even though both ultimately fall short of attention that is consistently and exclusively pulmonary. Given the masking experiments in Section 5.5.2, the near-chance COVID attention of the fine-tuned model is a reminder that its very high COVID recall (97.8%, Table 5.4) should not be assumed to reflect improved pulmonary diagnosis without further, dedicated causal testing on this specific model.

5.6 Comparison with the Simple CNN Baselines

Table 5.10 brings together the Simple CNN baselines of Section 4 and the transfer-learning models of this chapter on the common test split.

Table 5.10: Comparison of all principal models evaluated in this project (test split).

Model	Test Accuracy	Macro F1
Simple CNN, full image	81.96%	0.827
Simple CNN, lung-ROI	78.71%	0.774
Transfer, frozen EfficientNetB0	89.70%	0.907
Transfer, augmented (no flip)	88.38%	0.896
Transfer, fine-tuned	**92.36%**	**0.936**

Figure 5.7 shows the same comparison extended to every transfer-learning run trained in this chapter.

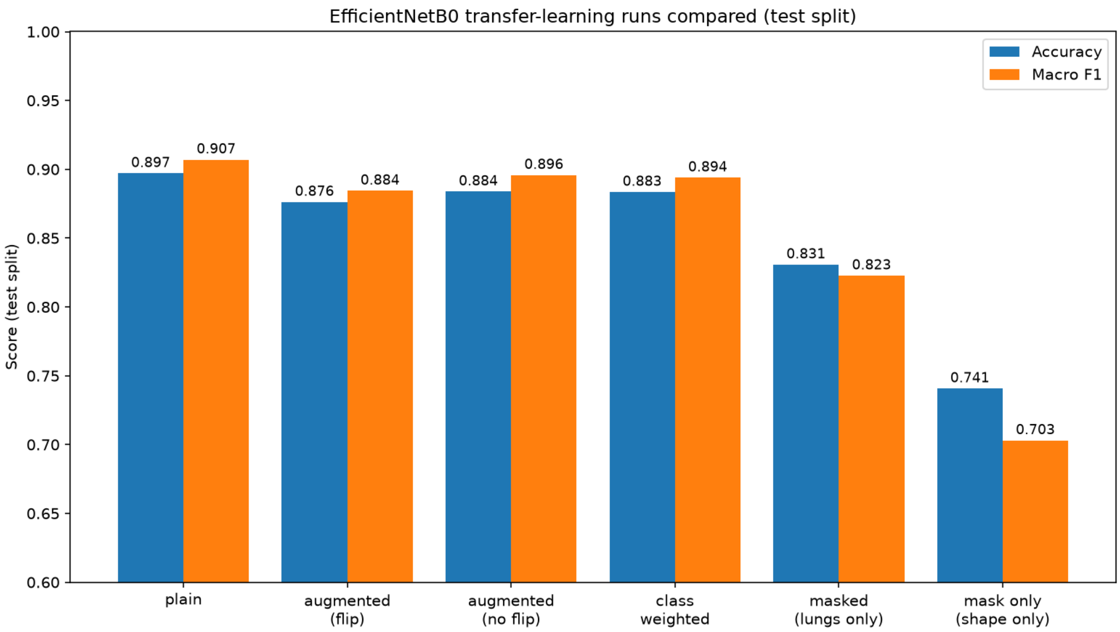


Figure 5.7: Test accuracy and macro F1 across all EfficientNetB0 transfer-learning runs compared in this chapter.

Three conclusions follow from this chapter, in line with the project's Step 3 objective of drawing scientific and business conclusions from the success or failure of the modelling phase.

First, transfer learning substantially outperforms training a comparable CNN from scratch on this dataset, and fine-tuning the backbone end-to-end gives the strongest result obtained anywhere in this project (92.36% accuracy, 0.936 macro F1). This is consistent with the dataset being too small, relative to the visual complexity of chest radiographs, for a from-scratch model to learn as effective a feature representation as one pretrained on a much larger, general-purpose image corpus.

Second, this headline performance is not fully attributable to pulmonary pathology. The interpretability analysis in Section 5.5 shows that a meaningful share of the frozen model's accuracy, and specifically its COVID recall, depends on non-anatomical cues in this Kaggle dataset rather than lung tissue itself. The fine-tuned model's near-chance COVID attention (Section 5.5.5) suggests this concern is not resolved simply by improving raw accuracy through fine-tuning, and may require dedicated causal testing analogous to Section 5.5.2 before it can be ruled out.

Third, and consequently, the headline accuracy of the best model in this project should not be read as evidence of clinical readiness. As with the CNN-from-scratch baselines, this remains an educational decision-support study rather than a validated diagnostic system, and any future external validation of this model line should specifically test for COVID-recall collapse on radiographs from sources not represented in this training dataset, since that is precisely the failure mode predicted by the background-leakage evidence in this chapter.

Chapter 6

Results

Chapter 7

Discussion

Chapter 8

Conclusion